

SUMMARY

3.0 Network and Internet Technology

3.1 What is Network?

3.1.2 Describe several uses of network.

	Network Use	Description
1	The Internet Applications	allows many networking applications to take place such as information retrieval, shopping, entertainment, accessing web and exchanging e-mail
2	Telephone Service	Provides telephone service to conventional landline phones or cellular network Television
3	Television and Radio Broadcasting	telecommunications network for distribution of television program and radio signal to public audience
4	Global Positioning System (GPS)	a satellite navigation system used to determine the ground position of an object
5	Applications Monitoring Systems	a networking technology to determine the current location or status of an object
6	Multimedia Networking	The used of networks to deliver multimedia content to devices on the home network
7	Videoconferencing	the use of networking technology to conduct real time, face to face meetings between individuals physically located in different places.
8	Collaborative Computing and Telecommuting	a networking technology with collaborative software tools to enable individuals to work together on documents and other project components from home or outside of office.
9	Telemedicine	the use of networking technology to provide medical information and services

3.1.4 Describe wired and wireless transmission media.

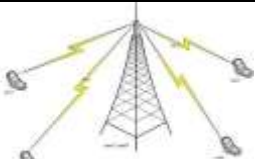
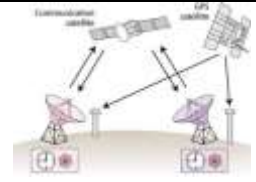
Transmission Media (Network Media)	
Wired	Wireless
Use wire, cable, and other tangible materials to send communications signals.	Use air or space (radio, microwave, and infrared signal) to send communication signals.

Wired Transmission Media

Twisted-pair cable	Coaxial cable	Fibre-optic cable
		

Wired Transmission Media	Characteristics	Usage
Twisted-pair cable	Twisted-pair cable consists of two separate insulated copper wires that are twisted together to reduce noise that can degrade communications.	- Used inside most homes for telephone communications. - Commonly used for a network such as a computer laboratory.
Coaxial cable	Coaxial cable consists of a single copper wire surrounded by a layer of insulating material, a woven/braided metal, and a plastic outer coating.	Used today in computer networks, for short run telephone transmission outside of the home, and cable television delivery.
Fibre-optic cable	The core of a fiber-optic cable consists of dozens or hundreds of thin strands of glass or plastic that use light to transmit signals.	Used for the high-speed backbone lines of a network, such as to connect networks housed in separate buildings or for the Internet infrastructure.

Wireless Transmission Media (Network Media)

Cellular radio Transmission	Microwave Transmission	Satellite Transmission	Infrared Transmission
			

Wireless Transmission Media	Characteristics	Usage
Cellular Radio Transmission	Radio signals are sent to and from cell phones via cellular towers.	widely used for mobile communications, specifically wireless modems, and cell phones.
Microwave Transmission	Uses high frequency radio signals that can send large quantities of data at high speed over long distances	<u>used in wide-open areas such as deserts or lakes, between buildings in a close geographic area, or to communicate with a satellite.</u> - Used in environments where <u>installing physical transmission media is difficult or impossible</u> and <u>where line-of-sight transmission is available.</u>
Satellite Transmission	Receives microwave signals from an earth-based station, amplifies (strengthens) the signals, and broadcasts the signals back over a wide area to any number of earth-based stations. These earth-based stations often are microwave stations. Other devices, such as smart phones and GPS receivers, also can function as earth-based stations	Used in remote areas.
Infrared transmission	Sends data as light rays over relatively short distances.	- Used for short range communication. - E.g., Mobile computers and devices, such as a mouse, printer, and smart phone

3.1.6 Describe the function of each communication device.

	Communication Device (Network Device)	Function
1	Network adapter (NIC)	• a network device used to connect a computer to a network. • <u>enables a computer or device</u> that does not have built-in networking capability <u>to access a network</u>.
2	Modem	• to connect a computer to a network over telephone lines. • converts data from a digital format into an analogue format and vice versa.
3	Switches/Hub	• provides a central point for cables in a network. • to share the information & resources by <u>connecting different network devices</u>, such as computers, printers, and servers, within a small business network. ◦ <i>receives data from many directions and then forwards it to one or more destinations.</i>
4	Routers	• <u>connects multiple networks</u> and allows the networked devices and users to access the internet. • Connects multiple computers & other routers.
5	Wireless Access Point	• allows computers and devices to transfer data wirelessly among themselves or to a wired network
6	Repeaters	• extend transmissions so that the signal can cover longer distances. • regenerate the signal over the same network before the signal becomes too weak or corrupted.

3.2 The Internet and World Wide Web

3.2.1 How to search effectively for information on the Internet and how to cite Internet sources properly.

- a) Describe types of search tools
(Search Engine-Keyword Search
Subject Directory - Directory Search)

Criteria	Search Tools	
	Search Engine	Subject Directory
Definition	software that finds information on the web related to a specific topic.	Classifies webpages in an organized set of categories and subcategories.
How to use	requires users to enter word or phrases which is known as search text (keyword)	User can locate webpage or information by clicking the links through different levels, moving from the general to the specific.
Best use	User has specific ideas on what to search by using keywords.	User doesn't have any precise idea on what to search and need to be guided by search category
Example	✓ Google Search ✓ Yahoo! Search ✓ Microsoft Bing ✓ Windows Live Search	✓ Google Directory ✓ Yahoo! Directory ✓ Look Smart Directory

b) Describe different search strategies.

	Search Strategies	Description	Example
1	Using phrases	• Use more than one keywords as search text. • Usually, allow use certain characters such as quotation mark (" ") to search the entire phrase together.	International Islamic University (60 308 280 hits) "International Islamic University" (1 205 772 hits)
2	Using Boolean operators	To further specify exactly what you want to find, Boolean operators such as AND, OR or NOT can be used in keyword searches.	Intel AND AMD microprocessor (hits would cover both processor only) Intel OR AMD microprocessor (hits would cover any of the processor) NOT Intel microprocessor (hits would not cover Intel processor)
3	Using field searches	Limit the search to a particular search characteristic (field) such as page title, URL, page text, top-level domain, or Website.	title: Halloween disaster (search Web pages containing the word Halloween disaster in the page title) url: Halloween disaster (search Web pages containing the word Halloween disaster in the URL) text: Halloween disaster (search Web pages containing the word Halloween disaster in the text of the page)

3.2.2 Describe types of web activities.

	Web Activities	Description	Example
1	Instant Messaging	a type of online chat allowing real-time text transmission over a network.	✓ WhatsApp ✓ Yahoo! Messenger ✓ Telegram ✓ Facebook Messenger
2	Text Messaging	the act of composing and sending electronic messages between two or more users of mobile devices or computers.	✓ Short Message Service (SMS) ✓ Multimedia Message Services (MMS)
3	Social Networking/social media	Social media are <u>interactive digital channels that facilitate the creation and sharing</u> of information, ideas, interests, and other forms of expression through virtual communities and networks. A social networking site is an <u>online platform</u> which <u>people use to build social networks or social relationships with other people</u> who share similar personal or career content, interests, activities, backgrounds, or real-life connections.	✓ Facebook/MySpace/Twitter (Allow users to posts information about themselves for others to read) ✓ Youtube (Video sharing site) ✓ Instagram/Tiktok (Photo and video sharing social network.) ✓ Flickr/Fotki (Photo sharing site) ✓ Pinterest (Allow user to share ideas and snippets from Webpages) ✓ Meetup (Connect people in specific geographic areas with common hobbies and interest)
4	E-commerce	The activity of electronically buying or selling product or services online or transmitting fund or data over the Internet	✓ Online Shopping – Lazada, Shopee ✓ Online Banking ✓ Online Auction ✓ Online Investing
5	Online Entertainment	Downloading, playing, and enjoying entertainment media such as games, music, and videos over the Internet	✓ Online Music – iTunes, Spotify ✓ Online TV, Videos and Movies – Netflix, Youtube video ✓ Online Games – Final Fantasy, Doom

6	Online News, Reference, and Information	News organization have Web sites that are updated to provide access to current local and world news. Reference sites are designed to provide user access to specific types of useful information.	✓ Online news – CNN, Al Jazeera, BBC ✓ Online reference – Map generator, weather forecast
7	Online Education and Writing	Online education/e-learning - the process of education that takes place over the Internet - Content such as schedule, assignment, notes, reference and questions are delivered online. Distance Learning - The way of learning remotely without face-to-face between students and instructor. - Offers student the flexibility of studying from anywhere and at any time over the Internet. Online Testing/ Assessment The online testing can facilitate quick report on candidate results and progress. Web-based Training (WBT) - Allow employee to be trained virtually either synchronously or asynchronously	E.g.: Learning Management System (LMS) such as Google Classroom, Distance Learning Microsoft Team E.g.: Video conferencing (Google Meet), Video Recording (Youtube) Kahoot!, Quizizz, Quizlet ✓ LinkedIn Learning ✓ Synchronously - Teleconferencing and webinars) ✓ Asynchronously – (e-learning courses and screencast)

3.2.3 Censorship and privacy and how they are related to Internet used.

b) Describe privacy.

Privacy (Definition)

Privacy is enclosing what information about individuals is available, how it is used and by whom.

	Type of Internet Privacy	Threat	Precautions/ how to overcome
1	Web Browsing Privacy	• Cookies • Spyware and Adware	Set medium high privacy setting in the browser to allow only certain regular cookies. Use anti-virus or Spyware and Adware remover to remove unwanted programs that can interrupt the computer system.
2	E-Mail Privacy	• unauthorized entities can intercept and read the e-mail. • Phishing attempts through emails.	✓ Encrypt sensitive e-mail. ✓ Secure confidential attachment such as using password. ✓ Always enable 2 steps verification ✓ Educate Users on Phishing Awareness ✓ Be caution when clicking on links or downloading attachments from unknown sources.

3.3 Latest issues of Internet

3.3.1 Discuss latest issues of Internet. (The Internet of Things (IoT), 4iR)

The 4th Industrial Revolution (4iR)

- conceptualizes rapid change to technology, industries, and societal patterns and processes in the 21st century due to increasing interconnectivity and smart automation.
- 4iR is changing how we live, work, and communicate using technologies like
 - ✓ artificial intelligence (AI),
 - ✓ autonomous vehicles (AV) and
 - ✓ the Internet of Things (IoT).

Benefits / Advantages	Risks / Disadvantages
New business opportunities and value creation	Job redeployment due to change in skills requirements
Efficiency and productivity gains	Irresponsible use and manipulation of technology
Transformation of 'dangerous, dirty and difficult (3D)' jobs	Erosion of trust in society due to increased privacy concerns and cyber threats
Improved quality of life	Widening inequality
Improved environmental quality	Deterioration of value and ethics
New job creation	Social interaction and societal wellbeing affected
Saving us money IoT can lead to cost savings through increased efficiency, reduced downtime, and optimized resource utilization.	Cost of Implementation can be expensive.
Providing us with better healthcare and other services	Security Concerns. It can lead to unauthorized access, data breaches, and potential manipulation of connected systems

Example of AI applied in everyday life:

- ✓ Chatbot- use to answer customer queries in e-Commerce platform.
- ✓ Google AI-Powered prediction – use to analyse the speed of traffic movement and report construction and accident in Google Map
- ✓ Facebook Face Recognition- allow the apps to highlight face and identify friends.
- ✓ Alexa- AI-powered personal assistant that accepts voice commands.

Usage of IoT in everyday life:

- 1) Home automation
- 2) Fitness systems
- 3) Smart home
- 4) Elder care
- 5) Medical and healthcare
- 6) Transportation
- 7) Agriculture
- 8) Product digitalization